



**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine

Rheumatologic Diseases Affecting the Muscle and Skin

Principles and Practice of Osteopathic Medicine I

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Session Objectives

At the conclusion of the session/course, the student should be able to:

- Describe the presentation and diagnostic tests used to diagnose Mixed Connective Tissue Disease (MCTD)
- Describe the presentation and diagnostic criteria used to diagnose dermatomyositis & polymyositis.
- Discuss the presentations of Vasculitis.
- Discuss the overlap of presentation with non-rheumatologic diseases.

What is Mixed Connective Tissue Disease (MCTD)?

- Patients with findings of more than one connective tissue disease
 - Example: patient with findings of both SLE and Systemic Sclerosis
- Most are ANA positive
- Key diagnostic feature is the production of autoantibodies to **U1 ribonucleoprotein (RNP)**
- Possible Pathogenesis: direct binding to endothelial cells / immune complex formation
- Controversy to diagnosis – area for much needed research
 - In some resources may be noted as undifferentiated autoimmune disease or a subtype of SLE or SSc

<https://www.ncbi.nlm.nih.gov/books/NBK542198/>

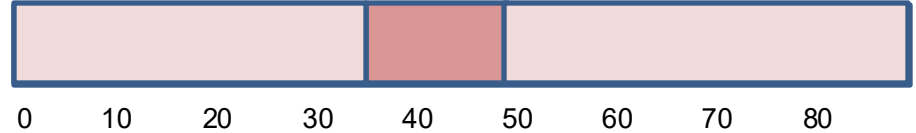


Epidemiology of MCTD

Who



Sex:
Female predominant



Mean Diagnosis 35-50



Race/Ethnicity
Based on Underlying Disorder

Criteria for diagnosis of MCTD

- **Must:**
 - Raynaud phenomenon, puffy digits, or hand edema
 - High titer + anti-U1-RNP
- **2 of the following:**
 - Synovitis, Myositis, leukopenia, esophageal dysmotility, pleuritis, pericarditis, or interstitial lung disease



Presentation and Diagnostic Challenges

- Non-specific presentation
 - Fatigue
 - Myalgias
 - Arthralgias
 - Low-grade fevers
- Typically, will be diagnosed with one connective tissue disease initially and then other disorders present later.
 - *Raynaud phenomenon common*



Source: Mosca M, Tani C, Neri C, Baldini C, Bombardieri S. Undifferentiated connective tissue diseases (UCTD). *Autoimmun Rev.* 2006 Nov;6(1):1-4. Epub 2006 Apr 19.
Review. PubMed PMID: 17110308.

Differential Diagnosis of MCTD

- Rheumatoid arthritis
- SLE
- Scleroderma
- Sjögren syndrome
- Polymyositis/Dermatomyositis
- Eosinophilic fasciitis
- Graft-versus-host disease



Sex:
2:1 F:M

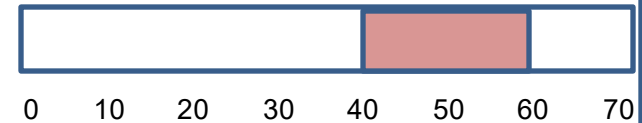
High rate of malignancy with DM:

- Ovarian
- Lung
- Pancreas
- Stomach
- Colon

Dermatomyositis

Path:

Cd4+ T lymphocyte
Damage perimysium
and blood vessels
Perifascicular
atrophy/necrosis



Peak Incidence 40-60 y/o

Peak Incidence 5-14 y/o

Polymyositis:

- Damages endomysium
- **Proximal b/l muscle weakness**
- Leg weakness occurs before UE
- No facial effects
- Painful
- Difficulty with swallowing

Dermatomyositis:

- Dark red malar rash
- Erythema on face, shoulders upper chest and back
- Periorbital edema:
- **heliotrope rash**
- **Gotttron papules**
- Nail-fold capillary dilation

Dermatomyositis and Polymyositis

Dermatomyositis:
Gottron Papules
Heliotrope rash
Nailfold Capillary
Dilation



Elevated CK
Muscle Biopsy
Special Tests:

- Anti-Jo-1 Ab most common
- Anti-Mi-2 Ab → DM
- Anti-SRP Ab → PM
- Anti-115/140 relates to risk of malignancy

Treatment

- Corticosteroids
- Methotrexate
- IVIG
- Rituximab
- Limit sun exposure

“Antisynthetase Syndrome

- Patient with findings of both PM and DM with findings similar to SLE arthritis

- Anti-synthetase Syndrome
- Immune-mediated necrotizing myopathy
- Sporadic inclusion body myositis

Myositis specific antibodies

Antibody	Clinical Association
Anti-Jo-1 and other antisynthetase antibodies	Polymyositis or dermatomyositis with interstitial lung disease, arthritis, mechanic's hands
Anti-Mi-2	Dermatomyositis with rash more than myositis
Anti-MDA5 (anti-CADM 140)	Dermatomyositis with rapidly progressive lung disease, cutaneous ulcers
Anti-155/140	Cancer-associated myositis
Anti-140	Juvenile dermatomyositis
Anti-SAE	Cancer-associated dermatomyositis, dermatomyositis with rapidly progressive lung disease
Anti-signal recognition particle	Severe, acute necrotizing myopathy
Anti-HMG CoA reductase	Necrotizing myopathy related to statin use



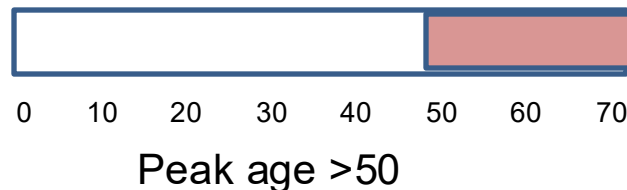
Polymyalgia Rheumatica



Sex:
2:1 F:M
Caucasian

T cell immune
reaction.
Inflammation
proximal joint
bursa and
tendons

- **Pain/stiffness:**
Proximal, neck,
shoulders, hips.
- Systemic- malaise,
fever, weight loss
- **No true weakness**
- Elevated **ESR** and
CRP. CK normal.
- Assoc. *Giant cell
Arteritis*
- *Vascular
pathogenesis?*



Treatment:
Steroids
methotrexate

Extended Differential Dx of Myopathies

Inflammatory	Drugs and toxins
Polymyositis	Illicit drugs - cocaine, heroin & Alcohol
Dermatomyositis	Corticosteroids
Inclusion body myositis	Other - colchicine, antimalarial drugs, HMG-CoA reductase inhibitors, penicillamine, zidovudine
Juvenile dermatomyositis	Infections
Vasculitis	Viral - influenza, parainfluenza, Coxsackie, HIV, cytomegalovirus, echovirus, adenovirus, Epstein-Barr virus
Overlap syndromes - lupus, scleroderma, rheumatoid arthritis, Sjögren's syndrome	Bacterial - pyomyositis, Lyme myositis
Rheumatoid arthritis, Sjögren's syndrome	Fungal
Endocrine disorders	Parasitic - trichinosis, toxoplasmosis
Hypothyroidism	Rhabdomyolysis
Cushing's syndrome (or exogenous steroid administration)	Crush trauma
Electrolyte disorders	Seizures
Hypokalemia	Exertion, especially with environmental heat illness
Hypophosphatemia	Vascular surgery
Hypocalcemia	Malignant hyperthermia
Hypernatremia or hyponatremia	Inherited myopathies
Metabolic myopathies	Acid maltase deficiency
Disorders of carbohydrate, lipid, and purine metabolism	Muscular dystrophy

Vasculitis: *inflammation of blood vessels*

Vessel Wall inflammation, can be systemic or limited to single organ/tissue

Classified as:

- Infectious
- Noninfectious

Noninfectious:

- Genetic predisposition > environmental trigger.
- Primary or secondary

Vasculitis Pathology

- Immune complex deposition- walls of vessels
 - Activate complement > neutrophils > lysosomal enzymes cause damage.
 - Examples: IgA vasculitis, Polyarteritis nodosa.
- Antineutrophil cytoplasmic antibodies (ANCA)
 - Target components of neutrophils, monocyte lysosomes, and endothelial cells
 - GPA and MPA
- Anti-endothelial cell antibodies
 - GPA, Kawasaki, Takayasu
- Autoreactive T cells.
 - Cytokine and macrophage activation

What

Vasculitis Overview

Large Vessel

- Giant Cell
- Takayasu arteritis

Medium Vessel

- Buerger
- Kawasaki
- Polyarteritis nodosa

Small Vessel

MPO/p-ANCA

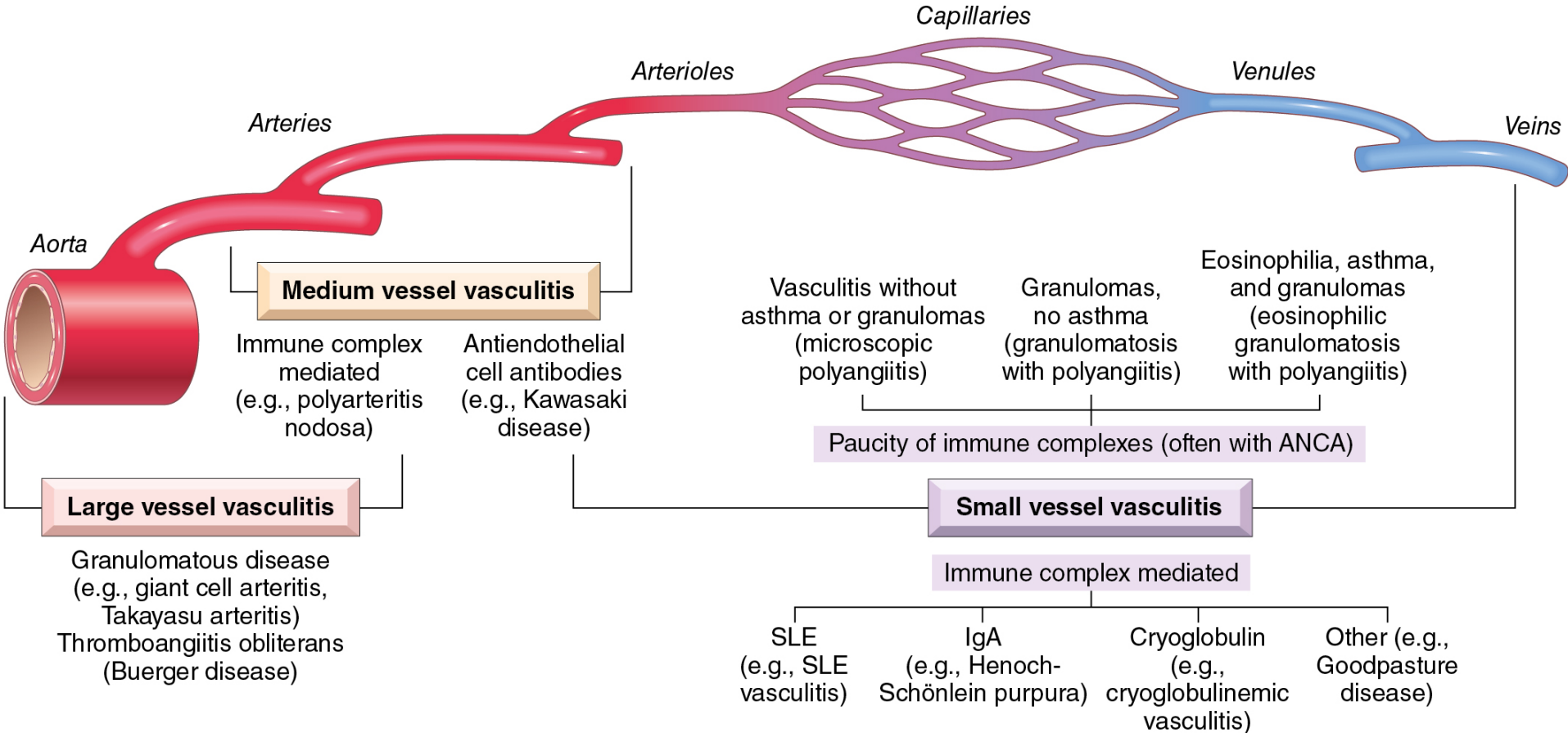
c-ANCA

All vessel

- Behcet syndrome

Vasculitis Overview

What



Richard N. Mitchell; Marc K. Halushka, 11, 441-478

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Vasculitis

- When to consider:
 - Systemic illness + ischemia otherwise unexplained
 - Elevated **ESR** and CRP
 - Biopsy
 - Combinations of renal, lung, and skin diseases (petechiae/purpura).

Large Vessel

Giant Cell Arteritis

- Granulomatous T-cell mediated
- Aorta and/or carotid, vertebral, and temporal arteries
- Age >50
- Symptoms: *headache, diplopia, vision loss, fever / weight loss / fatigue*
- Dx: Biopsy

Takayasu Arteritis

- Granulomatous T-cell mediated
- Age <50
- Symptoms: *ocular disturbances, weakening of upper ext. pulses.*
- Fibrous thickening of the aorta

Medium Vessel

Thromboangiitis obliterans Buerger Disease

- Associated with thrombosis and distal ischemia in extremities.
- Young males <35, heavy smokers
- Raynaud phenomenon, extreme pain, to gangrene

Polyarteritis nodosa

- Necrotizing
- Doesn't affect lung
- Affects: skin, kidney > heart > liver > gi tract
- Not assoc. with ANCA's
- 1/3 have hepatitis B.

Medium Vessel

Kawasaki disease

- Mucocutaneous lymph node syndrome
- Coronary arteries: aneurysm, thrombosis, rupture
- Infants and young children 80% <5 yo
- Acute, febrile, self-limited
- Delayed type hypersensitivity after viral infection.

Small Vessel

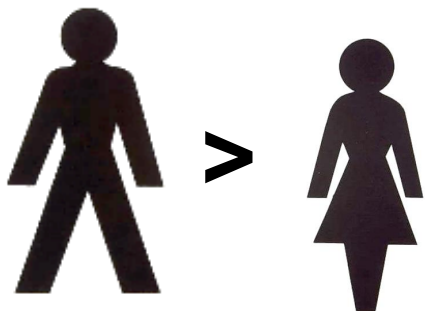
ANCA associated:

- Microscopic polyangiitis
 - Lungs and kidney
- Granulomatosis with polyangiitis “Wegener granulomatosis”
 - Nasopharynx, lungs, kidneys
- Eosinophilic granulomatosis with polyangiitis “Churg-Strauss syndrome”
 - Pulmonary, also heart and CNS

- Immune complex vasculitis
 - Deposits of immunoglobulin and or complement. Glomerulonephritis is frequent.
 - IgA vasculitis. “Henoch-Schonlein purpura”
 - children



Epidemiology of Behcet's Disease

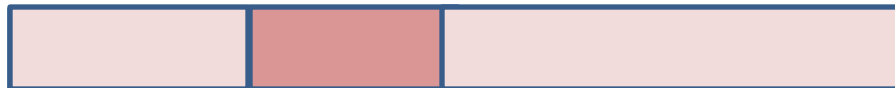


Sex:

More Common and
more severe in Males



HLA-B51/B5 allele
present in 57% of
people with Behcet



Can occur at any age but typically late
20s to 40s.
Avg: 30



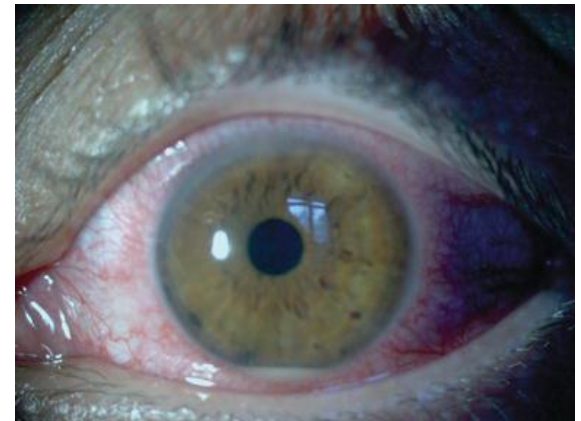
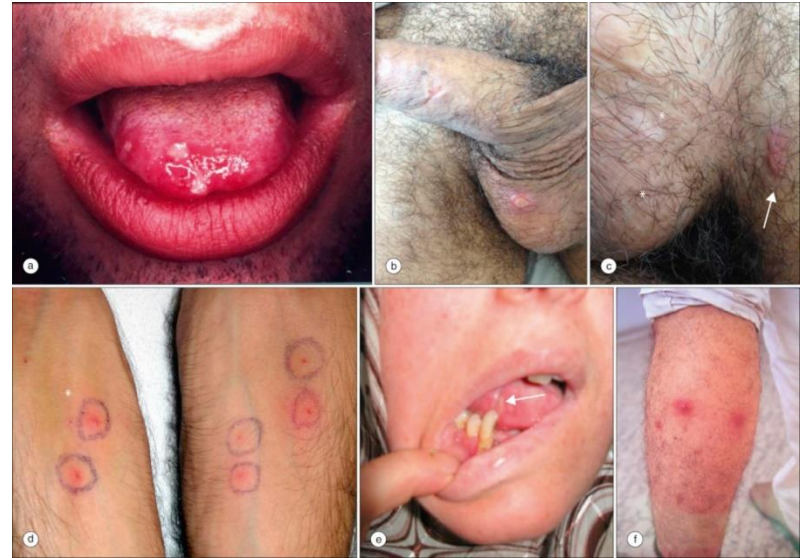
Most common in people from the
Middle East and Asia (Turkey and
Korean Peninsula)

Presentation of Behcet's Disease

Variable vessel vasculitis

Classic presentation is a patient presenting with reoccurring:

- **Oral** and genital ulcers
- Uveitis
- Joint and muscle pain
- Also affects: GI tract, CNS, and cardiovascular system



Source:

Behçet syndrome. Elsevier Point of Care Updated December 17, 2019.

Diagnosis of Behcet's Disease

Diagnosis is made clinically – laboratory testing is not required

option/not required: *Skin pathergy test**

Clinical criteria:

- Recurrent oral aphthosis
- Plus any two:
 - Ocular lesions
 - Genital aphthosis
 - Skin lesions
- Additional symptoms:
 - Joint pain
 - Neurologic symptoms
 - Vascular symptoms

Treatment of Behcet's Disease

- Treatment is scaled depending on symptom presentations
 - Corticosteroids (topical/oral/IV)
 - Colchicine
 - Immunosuppressants
 - TNF-alpha antagonists
- Prognosis: Males worse than females, no lifetime remission but some patients have decreased symptoms with age

Review

Treatment Overview of Vasculitis

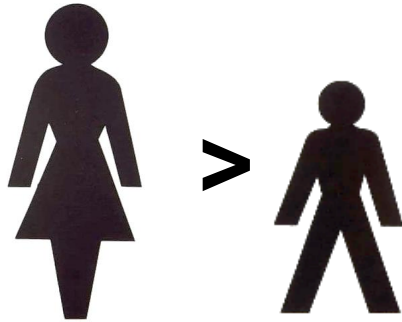
- Generally:
 - Prednisone
 - Methotrexate
 - Other considerations:
Tocilizumab/rituximab/cyclophosphamide
- Disease Specific treatments

Differentials:

Myopathy vs. Weakness

- Infectious vs. non-infectious
- Medications
 - Example: HMG CoA reductase inhibitor
 - (statins)
- Toxins/Alcohol
- Neurologic Pain
 - Fibromyalgia
- Congenital
- Endocrine disorders:
 - Example hypothyroidism

Fibromyalgia Considerations



Sex:

More commonly
diagnosed in Women
than Men



0 10 20 30 40 50 60 70 80

Can occur at any age but typical
onset late teens to early 40s



Disease of
exclusion – Not
auto-immune

Associated with chronic conditions,
including rheumatologic diseases.
Musculoskeletal pain
No association with autoantibodies

Review / Overview of tests

	RA	SLE	Sjögren	Diffuse Scleroderma	Limited scleroderma (CREST syndrome)	Polymyositis/ dermatomyositis
ANA	30–60	95–100	95	>95	>95	80
Anti-ds DNA	0–5	60	0	0	0	0
Rheumatoid Factor	70	20	75	30	30	33
Anti-Sm	0	10–25	0	0	0	0
Anti-SS-A	0–5	15–20	65	0	0	0
Anti-SS-B	0–2	5–20	65	0	0	0
Anti-SCL-70	0	0	0	33	20	0
Anti-Centromere	0	0	0	1	50	0
Anti-Jo-1	0	0	0	0	0	20–30
ANCA	0	0–1	0	0	0	0

Adapted from:

Maxine A. Papadakis, Stephen J. McPhee, Jennifer Bernstein+ Table 20-7.
Frequency (%) of autoantibodies in rheumatic diseases

Reminder of Other tests

- Acute-phase reactants
 - Serum erythrocyte sedimentation rate (ESR)
 - C-reactive protein (CRP)
- Creatine Kinase (CK)
- ANCAs (historically: c-ANCA / p-ANCA)

- Other References:
 - Access Medicine - Harrison's Principles of Internal Medicine, 21e - Chapter 365: Inflammatory Myopathies. Online Link:
 - Uptodate: Overview of the Management of Vasculitis in Adults. Merkel, Peter M.D.

https://www.uptodate.com/contents/overview-of-the-management-of-vasculitis-in-adults?search=vasculitis&source=search_result&selectedTitle=2~150&usage_type=default&display_rank=2

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